

DE424 Mini project

Find the Wumpus

	
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Up to this point in the module, we applied PGM concepts to relatively small problems, and I have given you a lot of guidance in the practical assignment for how to solve these problems. To end off the module, we will now be applying PGMs to a larger, more complex problem for which I will give you less guidance – tracking a wumpus (a type of monster) in a grid from inaccurate measurements.

1 Background

Object tracking: The *object tracking* task is to determine the trajectory of an object from inaccurate measurements of the object. The object measurements typically consists of two components: detecting an object in the scene, and measuring the location of the object. There are several types of errors that are typically found in the measurements; these include:

- *Noisy measurements* of the object's location; i.e., the measured object position differs from the actual object position
- *Missed detections* (false negatives); i.e., the object is present in the scene, but it is not picked up and no measurement is produced
- *Clutter measurements* (false positives); i.e., a measurement is generated that is not caused by the object, and the measured location is unrelated to the position of the object

Object tracking with these types of measurement errors are widespread. Possible examples include¹:

- Tracking an object (e.g. a ball) in a sequence of images: A common way to approach this problem is to build an *object detector* – an image processing algorithm that takes an image as input and produces the measured location of an object – and then to use the output of the detector as the input to an object tracking algorithm. All three of the types of errors mentioned above are typically present in the output of a detector.
- Tracking an object with radar sensors: Radars are notorious for generating loads of clutter measurements in some scenarios, and depending on the properties of the object, a radar sensor might struggle to detect an object in a scene.
- Do you remember the PGM model of the rhino poaching problem that we discussed at the start of the module?[1] A possible task related to this problem is to track a poacher in the Kruger National Park given indirect and inaccurate reports of the location of the poacher.

The wumpus grid world: A wumpus is a type of monster that I first encountered in an example in Russell and Norvig's book *Artificial Intelligence: A Modern Approach*[2], where it lives in a grid world and is used to illustrate the basics of reinforcement learning. This example was (apparently) based on the 1973 text-based computer game *Hunt the Wumpus*, where a player moves through a network of interconnected caves to find and shoot the wumpus, and the wumpus can be detected in a neighbouring cave by its smell. Anyway, the wumpus of this mini project is inspired by (but only very loosely based on) these examples. In the mini project, the wumpus wanders around in a grid world and is observed by inaccurate measurements.

¹When developing an engineering solution, it is always necessary to consider the ethical implications of the solution. For example, if you develop a powerful object tracking solution that works well under various types of measurement errors, then it could conceivably be used to infer the location of a supreme ruler of a country from a collection of uncertain intelligence reports. You will have to decide whether you can live with the consequences of all the possible uses of your solution, and you might have to put measures in place to restrict its use. Luckily, the solutions we develop in this mini project are fairly rudimentary and would not enable systems that can effectively be used for nefarious purposes (except from a wumpus' point of view), so we do not have to break our heads about it.

At each time step, the wumpus decides to move either one cell up, down, left or right, and it makes this decision randomly. It then tries to move accordingly: If the chosen cell is open, it moves into this square; otherwise, it remains in its current location (e.g., when it tries to move outside the boundary of the grid world). The movement of the wumpus for three time steps is illustrated for a 5×5 grid world in Figure 1. It is important to note that the

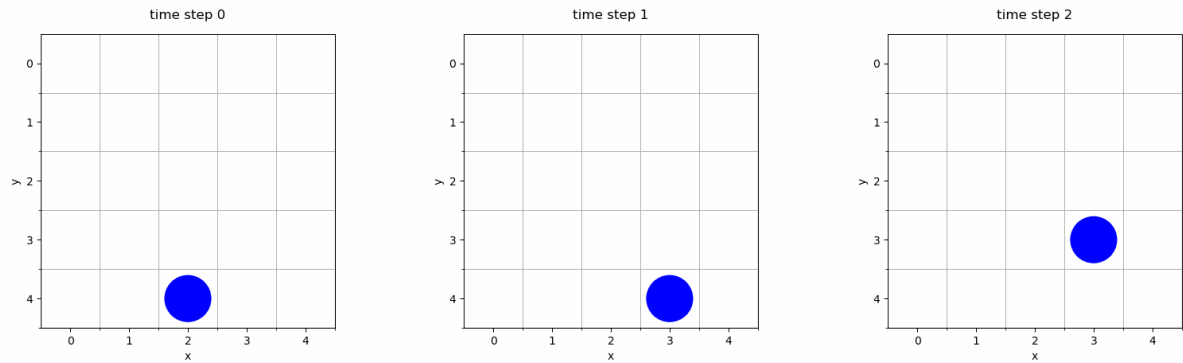


Figure 1: Illustration of the movement of the wumpus (blue circle) for three time steps in a 5×5 grid

wumpus position is latent; that is, it is never known exactly where the wumpus is.

At each time step, the location of the wumpus is inaccurately measured in the following manner:

- If the wumpus is located in a certain cell, that cell generates a detection with a (typically high) probability p_w . This cell therefore generates a missed detection (or false negative measurement) with a probability of $1 - p_w$.
- Every cell that does not contain the wumpus generates an incorrect detection with a (typically low) probability of p_c – this is a clutter measurement (or false positive measurement).

The measurements for three time steps (corresponding to the wumpus movement shown in Figure 1) are shown in Figure 2.

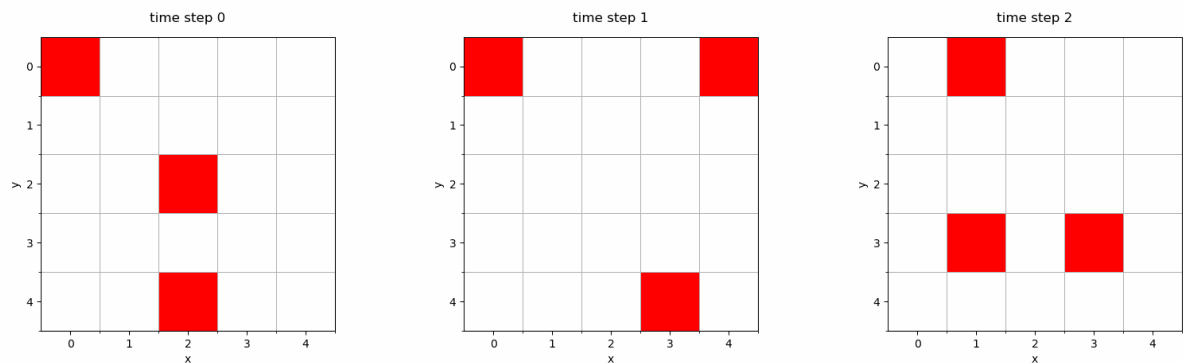


Figure 2: Illustration of detections generated for three time steps (corresponding to the wumpus movement shown in Figure 1) – cells that have generated a detections are indicated in red and the cells that have generated no detections are indicated in white

The detections generated by cells at each time step are known – these detections across all the time steps in a dataset are the evidence (or observations) that is available for inference. The detections for each time step are provided as a text file. For example, the detections corresponding to time step 0 in Figure 2 are given by a text file with the following contents:

```
1 0 0 0 0
0 0 0 0 0
0 0 1 0 0
0 0 0 0 0
0 0 1 0 0
```

2 Problem statement and objectives

The following datasets have been made available to you on STEMLearn:

- dataset1: A 5×5 grid; 10 time steps; known parameters – $p_w = 0.95$ and $p_c = 0.05$
- dataset2: A 20×20 grid; 20 time steps; known parameters – $p_w = 0.9$ and $p_c = 0.1$
- dataset3: A 10×20 grid; 20 time steps; unknown parameters
- dataset4: A 4×4 grid; 100 time steps; known parameters – $p_w = 0.95$ and $p_c = 0.05$. A number of cells are occupied (blocked), and if the wumpus attempts to move into these occupied cells, it will fail to do so (similarly to when it attempts to move outside of the bounds of the grid world). The number and location of the occupied cells are unknown. Any cell has a 25% chance of being occupied.
- dataset5: A 7×7 grid; 200 time steps; unknown parameters; grid world contains cells that are occupied (similar to dataset4). Any cell has a 10% chance of being occupied.

Your task for this mini project is to design, implement and evaluate a PGM-based solution that can take each of the datasets (i.e., a set of observed detections across a number of time steps), and track the wumpus (i.e., infer its position across all time steps) from these observations. This task can be broken up into the following 4 objectives:

Objective 1 – design and implement a model that models the wumpus world: A model of the wumpus grid world comprise two aspects: modelling the movement of the wumpus from one time step to the next, and modelling the observations for each time step. Your model should accurately capture the characteristics of the wumpus world, but also lead to accurate and efficient inference. There are many design choices you have to make during the modelling step, including (but not limited) to the following:

- What random variables (RVs) are you going to use, and how are they defined?
- Will you use a Bayes network or Markov network? And how will you connect up the RVs in your chosen type of PGM?
- Given your choice of RVs and PGM type and structure, how are you going to construct the factors for the model?
- Are you going to model any unknown parameters of the model as constant values, point estimates, or RVs?
- For the datasets with occupied cells, how will you represent these occupied cells, and how will you incorporate these representations into the model?

Objective 2 – construct a cluster graph, and choose an inference algorithm to track the wumpus: Given the model you set up as well as the evidence (the detections at each time step), you have to perform inference to determine the trajectory of the wumpus. For inference, you have to first construct a cluster graph and assign the factors of your model to clusters in the cluster graph. You have to design a cluster graph that is appropriate for your model, and which would lead to a good trade-off between accuracy and efficiency of inference.

Given a cluster graph design, you also have to choose one of the following inference algorithms (or you can try both and compare their performance):

- Marginal inference (i.e., using the belief-propagation algorithm)
- MAP inference (i.e., using the max-product algorithm)

For the implementation of the wumpus world model and setting up inference for it, you can choose to implement factors in `emdw` as early as possible in the process, or you could develop the inference solution “on paper” up to a point, and only then implement it in `emdw`. Notice which random variables would always be observed, and think whether you can exploit this knowledge when setting up inference.

Note that the idea is to use `emdw`’s built-in functionality as much as possible – do not implement any inference algorithm by hand if you would not gain anything by it.

Objective 3 – select or learn the parameters for your model: For some of the datasets, the detection parameters (p_w and p_c) are given to you, but for the other datasets, these parameters are unknown. For some of the datasets, there is also an unknown number of occupied cells in unknown locations, and you might want to use parameters to describe this part of the model. You have to decide how to view the parameters in your model (constants, point estimates, or distributions), as well as how to determine these parameters (making an informed choice for the parameter values, adjusting the parameter values by hand, or learning the parameters from data).

Objective 4 – evaluate your solution: Lastly, you should properly evaluate your solution (see Section 3 for more detail about the data available for evaluation). Evaluation of these types of solutions typically combine *qualitative* and *quantitative* evaluations. For qualitative evaluation, you would visually show the results for one or more scenarios; for this mini project, qualitative evaluation could amount to showing a number of time steps with the inferred wumpus location visualised in the grid world. Quantitative evaluation amounts to processing all your results and presenting it in numerical format. Since your solution can be thought of as a classifier (there are a finite number of cells in a grid, and each cell can be thought of as a class/category in a classifier), you can use any of several ways to evaluate a classifier.

3 Instructions

1. Study this whole assignment and make sure you understand the problem as well as all the requirements.
2. Reread the document about how to avoid academic misconduct in DE424 on STEMLearn to remind yourself how to complete the mini project in an acceptable manner. **For this mini-project, you must design the solution, implement the solution, and write the report by yourself. If you have been found guilty of academic misconduct in the mini project, then this could – amongst others – lead to you failing to satisfy the ECSA GAs, which will mean that you fail this module.**
3. Download and inspect the datasets made available to you.
4. Design a PGM-based wumpus tracking solution and implement the solution using `emdw`.
5. The ground truth trajectory for `dataset1` has been made available to you as a text file on STEMLearn. The first two lines of the text file are:

```
2 4
3 4
```

This represents the fact that the wumpus coordinates at time step 0 are $(x, y) = (2, 4)$, and the coordinates at time step 1 are $(x, y) = (3, 4)$, where the grid coordinates are defined as shown in Figure 1. By 13:00 on Thursday 14 May, the ground truth trajectories of the other datasets will be made available on STEMLearn. You may of course not use these ground truth trajectories in inference, but you can use them to evaluate the results of inference.

The coordinates of the occupied cells for `dataset4` and `dataset5` will also be made available by 13:00 on Thursday 14 May, as text files using the same format as the wumpus coordinates. You may again not use this ground truth information in inference, but you can use it to evaluate how accurately your inference algorithm has estimated the map.

6. Write a report using $\text{\LaTeX}$ or another document preparation solution (it may not be handwritten or contain handwritten parts); the report should include:
 - (a) At the top of the report, you should include a declaration that the report and the work presented in it are your own work. If you received any help or information from any source that contributed to the work or report, you should also mention the source and the extent of the help or information in the declaration.
 - (b) Next follows a section briefly describing how the ECSA graduate attributes (GAs) assessed in this module were met, as well as which parts of the report this is documented.
 - (c) Moving to the task itself, include the following:
 - i. A brief description of the problem
 - ii. A thorough, systematic description of your solution design. You should clearly explain the structure of your model and show all the factors, as well as state and motivate all your design choices. It should be clear from your report how you arrived at your final solution.

- iii. An overview description of the implementation of your solution in code. This description should be in the form of a discussion interspersed with code snippets, but should not simply be a listing of your code.
- iv. A brief description of how you processed the output of your solution, the presentation of your results, and an analysis thereof. You should report appropriate qualitative and quantitative results for all the datasets.

The length of your report is limited to **10 pages** (only the first 10 pages will be assessed; the rest will be ignored).

7. Submit your report (in PDF format) as well as all your code (C++ and other) on SUNLearn **before 23:59 on Friday 15 May 2026**. You should provide instructions (e.g., as a `README.txt` file) for how to combine and run the code. **Note that it must be possible to recreate the results you present in your report given your submitted code and instructions, and that the results in your report will be considered invalid otherwise.**

4 Tips

The mini project is somewhat different than the problems you've encountered so far in the module: the size of the model is larger, and it is a larger and more open-ended problem than the problems of the other practicals. Here are a few tips to help you to work efficiently through the project despite these new aspects.

4.1 Tips for the approach to the mini project

Don't be too ambitious: Avoid biting off more than you can chew. I suggest you aim to put the processing pipeline in place (in C++: reading detection data from file, converting it into factors, writing the result of inference to file; in e.g. Python: visualise the results, and get a simple solution design working as soon as possible (e.g., for `dataset1`). Once you have that in place, you can gradually extend your solution (e.g., by systematically working through `dataset2` to `dataset5`).

Plan: Make sure you don't run out of time – draw up a plan for the project and try to stick to it.

Work systematically: Gradually add functionality to your solution, and test the added functionality before continuing.

Write a good report: Leave enough time for yourself to write a good report. In fact, I suggest you start documenting your work in a report as soon as you start working on the project.

5 Assessment

Two aspects are assessed in the mini project: two ECSA graduate attributes (GAs) are assessed, and you will also receive a mark for the mini project that will contribute to your semester mark.

5.1 Assessment of ECSA graduate attributes (GAs)

The following two ECSA graduate attributes (GAs) are assessed in this module:

GA 1 – Problem solving: Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.

GA 2 – Application of scientific and engineering knowledge: Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialisation to develop solutions to complex engineering problems.

(green – assessed in this module; orange – partially assessed in this module; red – not assessed in this module)

The ECSA GAs are assessed in the mini project, and **the assessment is done based on your report** for the mini project. This means that an examiner or moderator must be convinced that you have satisfied both GAs based only on reading your report. Conversely, if you solved the problem well and applied scientific and engineering

knowledge appropriately in solving the problem, but these facts are not clear from your report, then you do not satisfy the GAs. **Note that if you do not satisfy the GAs, then you may not pass the module.**

You should add a section to your report that explicitly provides evidence of how and where these GAs were satisfied (as specified in Section 3). This section may (and should) be brief, but should discuss *all* aspects of the GAs being assessed. For example, GA1 partially assesses your ability to research literature as part of problem solving, which means that your report must mention and cite (at least) a few sources².

Please consult the module framework for a more detailed description of these two GAs as well as how they are assessed in this module.

5.2 General assessment criteria

Your report for the mini project will be assigned a mark out of 20, which will contribute to your semester mark. The aim of this aspect of the assessment is to judge the level to which you have mastered the material of this module and how well you have applied PGM concepts to this problem.

You must reach at least the minimum requirements for the mini project to achieve a pass mark. The minimum requirements can be summarised as follows:

- The most basic model design using appropriate PGM concepts from this module, which amounts to a model for a small (5×5) grid world with known detection parameters and no occupied cells.
- A correct implementation of the designed model using C++ and `emdw`
- Results that work reasonably well for at least `dataset1`.
- A report in which the model design, implementation and results are clear

Any significant aspect over and above the minimum requirements will contribute to a higher mark. This could include (but is not limited to):

- Estimating unknown parameters
- Tracking the wumpus in the presence of occupied cells, and inferring/estimating the map (i.e., which cells are occupied or empty)
- A more sophisticated or innovative model design, especially if it leads to better results
- The design, implementation and comparison of several different solution variations
- Excellent results across several datasets
- Testing of additional aspects (e.g. computational expense or convergence behaviour)
- Incorporating advanced or novel ideas
- An exceptionally clear and convincing report

Enjoy!

References

- [1] Hildegard Koen, Henk Roodt, and Alta De Waal. An expert-driven causal model of the rhino poaching problem. *Ecological Modelling*, 347:29–39, 2017.
- [2] Stuart Russell and Peter Norvig. *Artificial Intelligence: A Modern Approach*. Prentice Hall, 4 edition, 2022.

²Several of the prescribed resources of this module might help you to understand the problem, design a model, and implement a solution. You may use these prescribed resources in your research of literature – it is not necessary to do an extensive literature review.